



MITCOM/FITCOM

Steven Swain

Senior Airworthiness Inspector, (Regulatory Oversight Division)



Australian Government
Civil Aviation Safety Authority

MITCOM and FITCOM

Steven Swain
Senior Airworthiness Inspector, Regulatory Oversight

AEA Presentation



Session Overview

Topics Covered:

1. Definitions and Applicability
2. Scope and Limitations
3. Regulatory and Compliance Requirements
4. Common Practices





1. Welcome and Introduction
2. Technical Requirements
3. Interactive Features
4. Etiquette and Behaviour
5. Support and Assistance
6. Materials and Resources



Australian Government
Civil Aviation Safety Authority

MITCOM and FITCOM



MITCOM and FITCOM – Defining both

MITCOM - Manufacture in The Course of Maintenance Civil Aviation Regulations 1988 (CAR 1988).

Manufacture in the course of maintenance (MITCOM) applies to maintenance organisations holding an approval under regulation 30 of CAR and aircraft being maintained under Part 4 or Part 4A of CAR.

The term MITCOM is used to clearly distinguish between a part manufactured in the course of maintenance and a part manufactured under a CASR 21 manufacturing approval (APMA).

FITCOM - Fabrication in the course of maintenance, Civil Aviation Safety Regulations 1988 (CASR 1998).

Fabrication in the course of maintenance (FITCOM) applies to maintenance organisations holding an approval under Part 145 and the aircraft is being maintained under Part 42.

MITCOM and FITCOM – Defining both

MITCOM and FITCOM applies to Class III products (AC 21-03) that are used on an aircraft on which a maintenance organisation is carrying out maintenance. Products manufactured under MITCOM/ FITCOM are ineligible for sale or issue of an Authorised Release Certificate (ARC).

GUIDANCE MATERIAL FOR BOTH MITCOM AND FITCOM
AC 20-03 SECTION 2.3.7- MITCOM /FITCOM

Production of Parts (AC 21-08 v2.1)

Parts for a modification/repair design may be produced by the following means, **or a combination thereof** (see AC 21.27 for more information on manufacturing approvals):

- a) Production certificate
- b) MITCOM / FITCOM
- c) Australian Parts Manufacturing Approval (APMA) – see AC 21.16
- d) Australian Technical Standard Order authorisation (ATSOA) – see AC 21-601.

Class III Products

CASR 21.321 defines:

Class I – a complete type of certificated aircraft, aircraft engine or propeller

Class II – a major component of a Class I product i.e. wings, fuselage, empennage assemblies, landing gear, control surfaces, etc. the failure of which would jeopardise a Class I product.

Class III – Any part or component that is not Class I or Class II product and includes standard parts.

Examples: Bushes, sleeves, shims, skin panels, secondary structural elements, flexible hoses, rigid pipes, electrical cable looms, sheet metal panels, cable assemblies, etc.

Differences:

MITCOM

- NOT eligible for aircraft involved in CASR Part 42 Operations subject to agreement from CAMO.
- MITCOM is aircraft maintenance that needs to be certified in accordance with Regulations 42ZE or 42ZN of CAR 1988 as relevant.

FITCOM:

- FITCOM applies to Part 42 aircraft and Part 145 approved maintenance organisations.
- Some Part 145 organisations will be eligible for both MITCOM & FITCOM i.e. 145 AMO with CAR privileges.

Commonality:

- Can fabricate for known future maintenance needs of the organisation for authorised types.
- MITCOM / FITCOM products cannot be issued with an Authorised Release Certificate (ARC) - Form 1.
- Limited to Class III products
- Work Records need to be kept for at least 2 years (CAO 100.5 (5) and MOS 145.A.55 (c) 3.)
- Manufactured or Fabricated products cannot be produced for sale or transferred to other maintainers.



Australian Government
Civil Aviation Safety Authority

MITCOM



MITCOM - Overview

- Regulation 30 of CAR empowers CASA to grant a Certificate of Approval (COA) to an organisation authorising specified maintenance activities conditional upon having the necessary facilities, tools, trained & qualified personnel, etc.
- MITCOM is limited to Class 3 Products as defined in CASR 21.321
- Scope must be listed in procedures or in capability list.
- Must represent a specific need at a particular time e.g. when OEM item data not available.

MITCOM - Regulatory Basis- IAW CAR 42W(2)(b)(i):

Parts manufactured in the course of maintenance will need to comply with 42W(2)(b)(i) of CAR 1988. The MITCOM part must be either identical to the part it replaces, or conform to the approved design data for that part.



MITCOM – Compliance/ Limitations

- MITCOM capability / limitations must be listed in company procedures manual.
- Replacement parts for aircraft (or fleet) upon which currently conducting maintenance.
- Can't be on-sold or supplied to an outside organisation.
- **Does not qualify for an Authorised Release Certificate (ARC) - Form 1.**
- Parts must be identified & physically segregated during storage.
- Limited to aircraft being maintained under Part 4 or Part 4A of CAR 1988.

MITCOM - Airworthiness

MITCOM part must have no detrimental effect on the airworthiness of the aircraft or component.

Must be produced iaw approved data such as:

- Type Certificate
- Supplemental Type Certificate
- CASR Part 21M
- CAO 100.5 Paragraph 14
- AWB 02-045 Approval of Certain Maintenance data e.g. AC 43.13-1B Change 1 (where applicable).



MITCOM – Specifications

Must be produced to an approved engineering drawing:

- Part number
- Dimensions
- Materials
- Manufacturing processes (i.e. heat treatment etc.)

Sub-contracting allowable iaw approved procedures manual (where capability to do so has been demonstrated): – Welding, plating, machining, etc

MITCOM – Traceability

- Records of all parts made must be kept.
- All parts marked or tagged.
- Original Part Number + CoA Part Number/ID (IAW Procedures manual).
- Details included in maintenance records:
 - Pass these details to registered operator (who is then responsible for continuing airworthiness/defect reporting)
- Quality control inspections: Following manufacture & prior to installation.
- Clearly identified as a MITCOM part.
- Cannot be supplied to another maintenance organisation.



MITCOM – Considerations

- COA holder may manufacture for known future maintenance needs of the organisation for authorised capability (aircraft and component types).
- COA holder may subcontract certain specialist processes, subject to COA holder approved procedures and process controls.
- Parts manufactured under MITCOM continue to be the responsibility of the Registered Operator.
- ICA for manufactured part may require amendment to maintenance program (schedule).

MITCOM – Related data references

- Regulation 42W(2)(b)(i)
- CAAP 30-4(1) Certificate of Approval – Maintenance Organisation.
- CASR Part 21M
- CAO100.5 Paragraph 14
- CAAP 100.5
- AWB 02-045 Issue 9. Approval of certain types of maintenance data including FAA AC 43.13-1B. Change 1 or Repair of Aircraft.



Australian Government
Civil Aviation Safety Authority

FITCOM



FITCOM – Fabrication in the Course of Maintenance

Regulatory Basis:

CASR 42.435 Fitting parts fabricated by approved maintenance organisations – permission for subparagraph 42.420(5)(a)(ii):

For subparagraph 42.420(5)(a)(ii), the part may be fitted if:

- the part has been fabricated by an approved maintenance organisation; and
- the part is to be fitted to an aircraft on which the organisation is carrying out maintenance.

Regulatory and Guidance References

- CASR 1998
- CASR Part 145.005 -Purpose of Part
- CASR Part 145.015- Head of Power for Manual of Standards (MOS)
- MOS 145.A.43- Fabrication in the Course of Maintenance
- CASR Part 42.435-Fitting Parts fabricated by approved AMO
- CASR 42.420(5) (a) (ii)- Fitting other than Standard Parts
- CASR Part 145 Acceptable Means of Compliance (AMC) and Guidance Material (GM)
- AC 21-03.

FITCOM – Requirements

MOS 145.A.43 requirements:

- Aeronautical Product is fabricated and used IAW MOE procedures for aircraft or aeronautical products covered by its Approval Certificate.
- Must comply with all design data including, drawings, material, processes.
- Must comply with design data testing, inspections, testing.
- Where practical marked with Part number, Serial number and AMO identity IAW design data.

Maintain a record of

- Description, Part Number, Serial Number, A/C or AP type, Design data, and materials used.
- Installation details to A/C or AP and including the A/C records of ARC.
- Retain all appropriate records (2 years):
 - Annotate and cross reference aircraft maintenance records and FITCOM part record.



Questions?



Australian Government
Civil Aviation Safety Authority

MITCOM and FITCOM

Thank you for joining the session.

AEA Presentation